AMJAD YAGHI

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EDUCATION

University of British Columbia — Vancouver
Bachelor of Applied Science, Engineering Physics (Co-Op) — Expected May 2027

PROJECTS

Autonomous Badminton-Playing Robot (Capstone) ↗ | *C++20, OpenCV, Eigen, CMake*

- Designed a modular **C++20 static library** (pimpl, RAII, Eigen) powering a real-time stereo vision pipeline at ∼**100 FPS** with hardware-synchronized cameras over GPIO triggers
- Triangulated shuttlecock 3D position across 13 m via stereo rectification and OpenCV's `triangulatePoints`, with an offline ChArUco + court-transform calibration workflow
- Implemented an **Extended Kalman Filter** with hand-derived analytic Jacobian for nonlinear aerodynamic drag, plus RK4 forward simulation to predict landing point within ±**5 cm**

RL Robot-Arm Trajectory Tracking ↗ | *Python, MuJoCo, Gymnasium, Stable-Baselines3 (PPO)*

- Trained a **PPO policy** to track moving 3D targets with a simulated Franka Panda arm in **MuJoCo**; the policy outputs Cartesian lead targets, converted to joint torques via damped least-squares IK + PD control
- Beat the classical IK+PD baseline on a **200-seed paired evaluation** (mean error −**41%**, time-on-target **24%** → **73%**, lag **113** → **17 ms**), iterating v1 → v9 in a public build log

Autonomous Mario Kart Robot ↗ | *Arduino, STM32 (ARM), C++*

- Built a fully autonomous Arduino+STM32 Mario Kart robot using an FSM, **10 kHz** IR-beacon homing (via **convolution filtering**), and sonar edge detection for recovery; built a **zipline claw** that latched reliably to shortcut the course

STUDENT DESIGN TEAM

UBC SAE Aero Design — Vancouver
Payload Subteam, Sensing and Docking — May 2026 – Present

- Own IR beacon homing and self-docking for a rover that must find a landed aircraft, writing the **real-time ESP32 firmware** that interfaces with the team's **STM32H7** flight controller over UART, I2C, SPI or CAN
- Separated **three beacons** by carrier frequency and characterised detection out to **2 m** in room light, using a multiply-free **Goertzel filter** in bare-metal firmware at 81 samples per 27 ms steering update

EXPERIENCE

Algorithms, Optimization and Control Lab, UBC — Vancouver
Incoming Research Assistant — Oct. 2026

- Joining the team bringing the lab's Path Feasibility Governor (path planning with nonlinear model predictive control) from simulation onto an **autonomous indoor construction drone**, working on its RealSense depth perception

Generative Models.ai — Toronto
Software Engineer Intern — May 2025 – Aug. 2025

- Designed a **parallelized LLM blog-generation pipeline** that split outlines into per-heading prompts, enforced strict JSON schemas, and recombined validated sections into long-form drafts
- Implemented Django APIs and Temporal workflows to queue long-running analyses and **prevent request collisions**

SKILLS

Languages: C++, C, Python, SQL, MATLAB
Robotics & Tools: OpenCV, Eigen, ROS, Gazebo, MuJoCo, CMake, git, GitHub Actions, Docker, Linux
Embedded: firmware, microcontrollers (AVR, STM32, ESP32), bare metal, ADC/DSP, Analog Discovery 2